Boundary-Oriented Binary Building Segmentation Model With Two Scheme Learning for Aerial Images

Kyungsu Lee, Jun Hee Kim, *Member, IEEE*, Haeyun Lee, Juhum Park, *Member, IEEE*, Jihwan P. Choi, *Senior Member, IEEE*, and Jae Youn Hwang, *Member, IEEE*

***Abstract*— Various deep learning-based segmentation models have been developed to segment buildings in aerial images. However, the segmentation maps predicted by the conventional convolutional neural network-based methods cannot accurately determine the shapes and boundaries of segmented buildings. In this article, to improve the prediction accuracy for the boundaries and shapes of segmented buildings in aerial images, we propose the boundary-oriented binary building segmentation model (B3SM). To construct the B3SM for boundary-enhanced semantic segmentation, we present two-scheme learning (Schemes I and II), which uses the upsampling interpolation method (USIM) as a new operator and a boundary-oriented loss function (B-Loss). In Scheme I, a raw input image is processed and transformed into a presegmented map. In Scheme II, the presegmented map from Scheme I is transformed into a more fine-grained representation. To connect these two schemes, we use the USIM operator. In addition, the novel B-Loss function is implemented in B3SM to extract the features of the boundaries of buildings effectively. To perform quantitative evaluation of the shapes and boundaries of segmented buildings generated by B3SM, we develop a new metric called the boundary-oriented intersection over union (B-IoU). After evaluating the effectiveness of two-scheme learning, USIM, and B-Loss for building segmentation, we compare the performance of B3SM to those of other state-of-the-art methods using public and custom datasets. The experimental results demonstrate that the B3SM outperforms other state-of-the-art models, resulting in more accurate shapes and boundaries for segmented buildings in aerial images.**

***Index Terms*— Aerial images, boundary enhancement, deep learning, semantic segmentation.**

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I. Introduction

THE pixelwise labeling of satellite or aerial images is of great importance in many practical applications [1]. For example, the labels assigned to each pixel (e.g., ground, buildings, and roads) during the analysis of an aerial image can be utilized in various fields, including land cover mapping, traffic supervision, and forest cover tracking. In particular, frequent updates of information regarding building changes in cities under development are required for urban planning purposes [2]. However, because buildings exhibit significant variations in size, color, and angle, the semantic labeling of building geometries in aerial images is a challenging task [3]. Extracting buildings from aerial images is typically conducted using hand-digitizing processes for digital map generation [4], but the manual extraction of such features is laborious and time-consuming. Therefore, to overcome these issues, automatic methods, such as support vector machines [5] and artificial neural networks [6], have been developed to extract contextual features automatically during the image categorization process [7]. Recently, convolutional neural networks (CNNs) [8], which consist of series of convolutional filters, have exhibited excellent performance for the extraction of contextual features for the semantic segmentation of various types of images [9]–[12]. Although CNN-based studies [2], [13]–[17] have shown excellent potential in various fields, they still have limited capabilities that the output segmentation maps from CNNs are typically too coarse, and their boundaries are not perfectly aligned with the corresponding ground truth. These issues are caused by the tradeoff between detection and localization in CNNs [18].

The accurate shapes and boundaries of predicted buildings are crucial for a digital map generation from satellite or aerial images in urban planning. However, during the feature extraction process, feature resolutions are reduced by pooling layers and the (de)convolutional striding, and feature maps with coarse resolutions also destroy detailed spatial information, resulting in poor boundary or edge detection. These issues are crucial for an aerial image segmentation because many small objects generate misclassifications at boundaries, hampering the overall performance of a system. Several types of architectures customized for aerial images have been proposed to compensate for the loss of spatial information [18]–[22]. However, although previous methods have provided improved

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segmentation results, the segmentation maps predicted by most methods still contain imperfect shapes and boundaries of segmented buildings. Therefore, a novel deep learning architecture should be developed to achieve better shapes and boundaries of segmented buildings.

In this study, we propose a novel deep learning model, the boundary-oriented binary building segmentation model (B3SM). The model is based on two-scheme learning to achieve accurate identification of shapes and boundaries of segmented buildings from aerial images. While constructing the B3SM, we developed a novel operator, called an upsampling interpolation method (USIM) and a boundary-oriented loss function (B-Loss); this enabled an improvement in the recognition of the shape and boundaries of segmented buildings. Once the novel functionalities of two-scheme learning were demonstrated, the effectiveness of USIM and B-Loss were also evaluated by examining the segmentation of buildings from an aerial image. The effectiveness of the BS3M, in terms of improving the shapes and boundaries of segmented buildings, has been evaluated quantitatively using intersection over union (IoU) and boundary-oriented IoU (B-IoU). The B-IoU was designed to evaluate quantitatively the shape and boundaries of segmented buildings. Finally, the B3SM, which consists of two schemes, USIM and B-Loss, is compared to other state-of-the-art models, including Adapted-FCN [23], U-NetPPL [14], MC-FCN [2], and MS-CDN [16], in terms of IoU and B-IoU. The results demonstrate that the B3SM outperforms other state-of-the-art models for the semantic segmentation of buildings in aerial images.

The main contributions of this article can be summarized as follows.

1) The novel B-Loss function is proposed for enhanced optimization of deep learning models, resulting in more accurate shapes of segmented buildings.
2) We also propose a novel evaluation metric called B-IoU to evaluate the predicted boundaries and shapes of buildings because conventional metrics may not be suitable for quantitative evaluation of the boundaries and shapes of predicted segmentation maps. The B-IoU metric assesses shape and boundary accuracy.
3) The B3SM with the USIM, two-scheme learning, and B-Loss exhibits improved performance with an IoU of 76.08% for the semantic segmentation of buildings. In addition, the B3SM exhibits a B-IoU of 78.16%, which is 8.42% greater on average than the values of the compared state-of-the-art models, indicating significant improvement in terms of the shape prediction of buildings during building segmentation tasks.

The rest of this article is organized as follows. In Section II, the related studies of this article are discussed. In Section III, data sources for aerial images are discussed. In Section IV, B3SM, two-scheme learning, B-IoU, and USIM are discussed in detail. Experiments and results are presented in Section V. Finally, the conclusion and our contributions are summarized in Section VI.

II. Related Work

A. Segmentation in an Aerial Image

Recently, many advanced state-of-the-art models have been proposed with the modification of their architecture to extract various object features from an aerial image. Among them, Kim *et al.* [14] suggested UNetPPL, which is a U-Net-based architecture [13], to achieve the extraction of multi-scale features by applying multiple pyramid pooling layers [15]. For a subdecimeter aerial image segmentation, Volpi and Tuia [24] utilized the downsampling and upsampling architecture for an improved geometric accurate of predictions. In addition, Yue *et al.* [25] devised novel adaptive layers, denoted as Tree-UNet, to improve the performance of a deep learning model for building segmentation using subdecimeter aerial images. Doi and Iwasaki [26] applied a focal loss [27] to extract focused features in vague aerial images. Maggiori *et al.* [28] analyzed an in-depth architecture of CNNs and then derived a CNN framework adapting to the semantic labeling problems. Moreover, the deep learning network with feature-oriented layers has been developed to extract more precise objects in an aerial image using hyperspectral data [11], [12]. Various iterative enhancement algorithms using multitasks have been proposed to sharpen coarse CNN outputs. Maggiori *et al.* [18] proposed the recurrent neural network for the segmentation of aerial images. Mou and Zhu [19] also derived the fusion of multilevel feature maps for enhanced segmentation of aerial images. Volpi and Tuia [29] proposed multitask learning while combining bottom-up information with top-down spatial regularization encoded by a conditional random field model, thus providing better regularization. Beyond the supervised learning for the segmentation of aerial images, unsupervised learning methods have been developed to avoid the hand-digitizing labeling tasks. Kemker *et al.* [30] utilized the synthetic DIRSIG data for optimization of a deep learning network, and the network was then fine-tuned using real geographic images as semisupervised learning. For unsupervised segmentation tasks, Saha *et al.* [31] devised the novel loss function that allows the segmentation of multiple objects from individual images using established corresponding distinct multitemporal segments, as well as a joint segmentation method.

B. Boundary Enhancements for Segmentation Tasks

Several studies have demonstrated methods for enhancing the boundary detection of objects in aerial images [32]–[34]. These studies primarily focused on segmenting the boundaries of objects in aerial images and improved the evaluation scores. Marmanis *et al.* [32] showed that deep learning-based networks were able to generate multiscale boundary predictions using a set of scale dependent class boundary classifiers. The multiscale boundary predictions were particularly improved by averaging the individual class probability predictions. The boundary-detection network, denoted as holistically nested edge detection (HED-H), was proposed. The main purpose of the proposed network is to extract the boundary-oriented features from multilevel feature maps. The extracted boundary-oriented features are utilized to precisely segment objects in aerial images. An important feature

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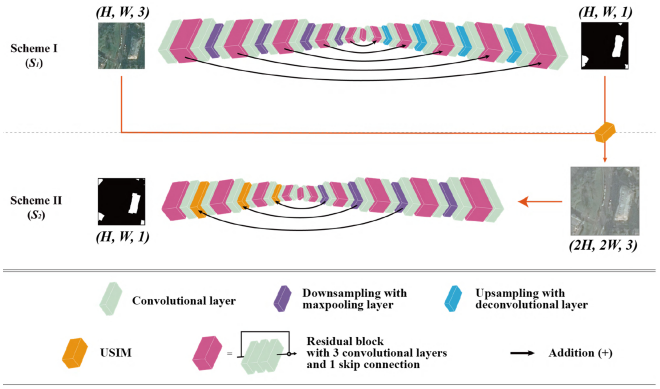


Fig. 1. Detailed structure of B3SM of which the baseline architecture is a modified FusionNet. Since the size of an image for Scheme II is two times larger than that of a raw input image, three USIMs are implemented in the decoder layer, whereas four downsampling operations are implemented in Scheme I.

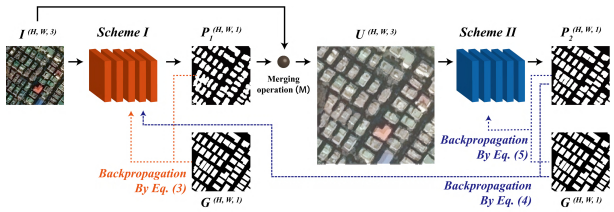


Fig. 2. Pipeline of the two-scheme learning architecture. With the two-scheme learning, the final predicted segmentation map, P_2, is generated. The trainable variables in each scheme are optimized using loss functions. The trainable variables in the first scheme, $\theta(S_1)$, are optimized by (3) and (4) based on P_1 and P_2, respectively. On the contrary, the trainable variables in the second scheme, $\theta(S_2)$, are optimized by (5) based on P_2.

of HED-H is the multiscale prediction in conjunction with deep supervision. The multiscale predictions are combined into a final boundary probability map. In this article, it is addressed that, even though HED-H has no explicit mechanism to enforce closed contours, HED-H tends to generate closed contours that are important to support a segmentation task by learning from global image context. In addition, new metrics were proposed to evaluate the boundaries of segmented objects in the predicted segmentation map. For example, Fernandez-Moral *et al.* [33] proposed the new metric, denoted as boundary Jaccard (BJ), to evaluate the boundary accuracy. The main function of BJ is to perform the boundary-to-object measurement, which evaluates the continuous boundary distance. Therefore, BJ is to compute the distance from the boundary binary map to the binary map of the predicted segmentation.

III. Methods

In this section, the architecture of B3SM and its subcomponents are demonstrated. Fig. 1 illustrates the pipeline of the proposed B3SM. The B3SM uses a modified FusionNet as baseline architecture for two-scheme learning. For the B3SM, we developed B-Loss to derive more accurate shapes for segmented buildings in aerial images. In addition, the USIM was designed as a connection between the first (S_1) and second schemes (S_2), as well as the decoder of the second scheme. In the following, the key subparts of B3SM are discussed in detail: 1) two-scheme learning; 2) USIM; and 3) B-Loss.

A. Two-Scheme Learning

We construct CNN-based two-scheme learning architecture to extract the features that are more associated with boundaries and shapes of buildings in aerial images. The B3SM with two-scheme learning is defined as

$$\begin{aligned}&\text{B3SM}\left(I^{(H,W,3)};\Theta\right)\\&\quad:=f_{S_2}\left(\mathcal{M}\left(I^{(H,W,3)},f_{S_1}\left(I^{(H,W,3)};\Theta(S_1)\right)\right);\Theta(S_2)\right).\end{aligned}\tag{1}$$

The mathematical symbols are listed in Table I. Each scheme (S_1 and S_2) has different deep learning architectures with trainable variables of convolutional operations and loss functions. A raw input image for the first scheme (S_1) of B3SM is defined as $I^{(H,W,3)}$ since the number of channels (C) of an RGB input image is 3. In addition, a predicted segmentation map, which is the output of the second scheme (S_2), and a ground truth, which is the correct segmentation map corresponding to an input image, are defined as $P^{(H,W,1)}$ and $G^{(H,W,1)}$, respectively.

By the definition of a convolutional function (f_S) in Table I, the two-scheme architecture of B3SM is defined as follows:

$$\begin{aligned}S_1:\ P_{S_1}^{(H,W,1)}&=f_{S_1}\left(I^{(H,W,3)}\right)\\S_2:\ P_{S_2}^{(H,W,1)}&=f_{S_2}\left(\mathcal{M}\left(I^{(H,W,3)},P_{S_1}^{(H,W,1)}\right)\right)\\&=f_{S_2}\left(\mathcal{M}\left(I^{(H,W,3)},f_{S_1}\left(I^{(H,W,3)}\right)\right)\right).\end{aligned}\tag{2}$$

Note that this two-scheme pipeline is not the retraining of the same architecture but optimizes nonshared variables while utilizing different loss functions. S_1 behaves as a supplementary training scheme for extracting the boundaries of buildings in aerial images, whereas S_2 acts as the main training scheme. In S_1, the BCE-loss and B-Loss functions optimize the set of trainable variables $\theta(S_1)$ for the prepredicted segmentation map ($P_{S_1}^{(H,W,1)}$) and corresponding ground truth ($G^{(H,W,1)}$) via backpropagation as follows:

$$\underset{\theta}{\text{minimize}}\ \mathcal{L}\left(P_{S_1}^{(H,W,1)},G^{(H,W,1)};\theta(S_1)\right).\tag{3}$$

In contrast, in the second scheme (S_2), the inputs, generated from a merging and upsampling operation ($\mathcal{M}$), are utilized to optimize $\theta(S_2)$. The merged image, generated by the operation ($\mathcal{M}$) from the raw input image ($I^{(H,W,3)}$) and prepredicted segmentation map ($P_{S_1}^{(H,W,1)}$) in S_1, is fed into the second function (f_{S_2}) and the corresponding CNN architecture in S_2. As the predicted final segmentation map ($P_{S_2}^{(H,W,1)}$) is generated by the function (f_{S_2}), the gradient of the trainable variables of $\theta(S_2)$ can be calculated via the optimization of the loss function with $P_{S_2}^{(H,W,1)}$ and ground truth (G). Similarly, as the predicted final segmentation map ($P_{S_2}^{(H,W,1)}$) is generated by the composite function of f_{S_1}, f_{S_2}, and $\mathcal{M}$ in (2), the gradient of trainable variables of $\theta(S_1)$ can be calculated using the loss function with $P_{S_2}^{(H,W,1)}$ and G. Therefore, the gradient of two sets of trainable variables ($\theta(S_1)$ and $\theta(S_2)$) can be optimized

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TABLE I
MATHEMATICAL SYMBOLS AND THEIR DEFINITIONS

Symbols	Definitions
$I^{(H,W,C)}$	Image of 3D matrix with the height(H), width(W), and channel(C).
S_i	The ith scheme. Here, S_1 and S_2 are defined.
f_S	The function which matches deep learning architecture in the scheme(S) as $f_S : M^{(H_1,W_1,C_1)} \rightarrow M^{(H_2,W_2,C_2)}$
$\theta(S)$, $\Theta(S)$	The set of trainable variables and fully optimized variables of functions in the scheme(S).
$\mathcal{L}(P^{(h,w,1)}, G^{(h,w,1)}; \theta(S))$	The loss function($\mathcal{L}$) with prediction(P), ground truth(G), and target variables(θ) to be optimized.
Combine	The function that merges two images of I_1 and I_2 as one output image of I_3. $combine(I_1^{(H,W,C)}, I_2^{(H,W,C)}) = I_3^{(H,W,C)}$
Upscale	The function that doubles the size of one image of I_4 as an output image of I_5. $upscale(I_4^{(H,W,C)}) = I_5^{(2H,2W,C)}$
Merge ($\mathcal{M}$)	The function that upsamples and combines I_6 and I_7 as one image of I_8. *USIM* is a kind of a *Merge* operator. $\mathcal{M}(I_6^{(H,W,C)}, I_7^{(H,W,C)}) = I_8^{(2H,2W,C)}$

Algorithm 1 USIM Operator Procedure

Input: $I_1^{(H,W,C)}$, $I_2^{(H,W,C)}$
Output: $U^{(2H,2W,C)}$

For h in {1, 2, …, H} **do**
  For w in {1, 2, …, W} **do**
    # fill pixel values with I_1
    For c in {1, 2, …, C} **do**
      $U(2*h, 2*w, c) = I_1(h, w, c)$
      $U(2*h+1, 2*w+1, c) = I_1(h, w, c)$
    end For
    # fill pixel values with I_2
    For c in {1, 2, …, C} **do**
      $U(2*h+1, 2*w, c) = I_2(h, w, c)$
      $U(2*h, 2*w+1, c) = I_2(h, w, c)$
    end For
  end For
end For

as follows:

$$\underset{\theta}{\text{minimize}}\ \mathcal{L}\left(P_{S_2}^{(H,W,1)}, G^{(H,W,1)}; \theta(S_1)\right) \tag{4}$$

$$\underset{\theta}{\text{minimize}}\ \mathcal{L}\left(P_{S_2}^{(H,W,1)}, G^{(H,W,1)}; \theta(S_2)\right). \tag{5}$$

In the second scheme, the IoU-Loss, in terms of (4) and (5), is utilized to optimize the B3SM for generating accurate shapes of buildings. The prediction and optimization processes are illustrated in Fig. 2 with the corresponding equations.

B. USIM: Boundary-Oriented Feature Extraction Module

In this section, the USIM is described in detail. The USIM is an operator for extracting features from the boundaries of buildings in aerial images. Algorithm 1 summarizes the pseudocode of the USIM operator. The USIM operator takes two input images and generates one merged image $U^{(2H,2W,C)}$ by merging and upsampling the input images as follows:

$$\mathcal{M}\left(I_1{}^{(H,W,C)}, I_2{}^{(H,W,C)}\right) = U^{(2H,2W,C)} \tag{6}$$

where I_1 and I_2 are the input images, U is the output image of the USIM, and H, W, and C are the height, width, and channel of each image, respectively. The merged image is two times larger than the input images but preserves the original pixel values of the input images. The USIM is implemented between schemes S_1 and S_2 in the proposed deep learning architecture. The predicted segmentation map generated by S_1 and the raw input image are merged by USIM, and then, the merged image becomes the input for S_2. Therefore, the function f_{S_2} in the second scheme uses the USIM to leverage the boundary-oriented features extracted in S_1.

C. Training B3SM

1) General Loss Functions: For building segmentation, two classes are defined as Class 0 and Class 1. Class 0 indicates the background, which includes all objects other than buildings, whereas Class 1 indicates buildings. During the minimization of the loss function, the element g_{hwc} of the ground truth $G^{(H,W,1)}$ is binarized as zero or one, and the element p_{hwc} of the prediction $P_S^{(H,W,1)}$ is set to continuous values between zero and one by a softmax layer as follows:

$$\begin{cases} g_{hwc} \in \{0, 1\} \\ p_{hwc} \in [0, 1]. \end{cases} \tag{7}$$

Since the elements of $P_S^{(H,W,1)}$ are linear probabilities, the gradients, ∇_θ, of trainable variables $\theta(S)$ can be calculated and then used to optimize the trainable variables. In this study, BCE-Loss was used in the first training scheme S_1 to generate prepredicted segmentation maps. The definition of the binary

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cross-entropy loss function is given as follows:

$$\mathcal{L}_{\mathrm{BCE}}(P, G; \theta) = -\nabla_\theta \frac{1}{n} \sum_{i=1}^{n} \sum_{h,w,c}^{H,W,C} \{g_{hwc} \log p_{hwc} + (1 - g_{hwc}) \log (1 - p_{hwc})\} \tag{8}$$

where g_{hwc} and p_{hwc} are the elements in G and P, respectively, and θ indicates the trainable variable. A stochastic gradient descent (SGD) method with a minibatch size of n is applied to minimize the loss function.

2) Area-Oriented Loss Function: Conventional loss functions only depend on likelihood but do not consider the shapes of buildings during the training of deep learning architectures [35]. Therefore, optimization using likelihood-based training is not suitable for boundary-oriented feature extraction from raw input images. In contrast, IoU-Loss is an area-oriented loss function [35]. In the quantitative analysis of binary classification, IoU values are used as measures of prediction accuracy. Therefore, IoU is generally utilized to evaluate the performance of deep learning models for semantic segmentation tasks. Because IoU-Loss optimizes trainable variables to increase the intersection area between the ground truth and the predicted segmentation map, predictions focus on intersection areas as given follows:

$$\mathcal{L}_{\mathrm{IoU}}(P, G; \theta) = \nabla_\theta \frac{1}{n} \sum_{i=1}^{n} \sum_{h,w,c}^{H,W,C} \left\{ 1 - \frac{g_{hwc} * p_{hwc}}{(g_{hwc} + p_{hwc}) - g_{hwc} * p_{hwc}} \right\} \tag{9}$$

where g_{hwc} and p_{hwc} are the elements in G and P, respectively, and they are real numbers as in (8). In addition, * and + indicate multiplication and addition operators, respectively. When the predicted segmentation map is the holistically same as the ground truth ($G = P$), IoU-Loss becomes zero since $g^2 = g$ such that $g * p = g^2 = g$, and then, the denominator and the numerator become the same. On the other hands, when the predicted segmentation map differs from the ground truth, IoU-Loss becomes larger.

3) Boundary-Oriented Loss Function: We designed B-Loss to achieve more accurate shapes for buildings in the segmentation of aerial images. Since the previously proposed optimization methods merely joined the boundary-extracted features regardless of the accurate extraction of the boundaries of objects [32] or measured the boundary similarity regardless of the scale of the objects, accurate predictions of the boundaries could not be performed [33]. Therefore, we designed the B-Loss function that allows pixel values near boundaries to affect the optimization of trainable variables during the training of deep learning architecture. To develop this loss function, we adopted the characteristics of a linear function (F) as follows:

$$P_S^{(H,W,1)} \cong G^{(H,W,1)} \Longrightarrow F\left(P_S^{(H,W,1)}\right) \cong F\left(G^{(H,W,1)}\right). \tag{10}$$

Equation (10) represents that $F(P_S^{(H,W,1)})$ becomes approximately equal to $F(G^{(H,W,1)})$ when $P_S^{(H,W,1)}$ is approximately equal to $G^{(H,W,1)}$. In this case, the B-Loss can be defined with the simple edge detection function as follows:

$$\mathcal{L}_{\text{B-Loss}}\left(P_S^{(H,W,1)}, G^{(H,W,1)}, \theta(S_1)\right) := \mathcal{L}_{\mathrm{IoU}}\left(F_{\mathrm{edge}}\left(P_S^{(H,W,1)}\right), F_{\mathrm{edge}}\left(G^{(H,W,1)}\right); \theta(S)\right). \tag{11}$$

The edge detection function F_{edge} for segmenting buildings is here defined as follows:

$$F_{\mathrm{edge}}(I) := F_d\left(F_{\mathrm{canny}}(F_e(I))\right) \tag{12}$$

where F_d, F_{canny}, and F_e indicate the Dilation, Canny Edge, and Erosion functions, respectively. In the previous study [36], the relation between IoU and IoU-Loss is defined as follows:

$$\mathcal{L}_{\mathrm{IoU}} = 1 - \mathrm{IoU}'(P, G). \tag{13}$$

Therefore, based on (11) and (13), the relation between B-Loss and B-IoU can be defined as follows:

$$\begin{aligned} \mathcal{L}_{\text{B-Loss}}(P, G) &= \mathcal{L}_{\mathrm{IoU}}\left(F_{\mathrm{edge}}(P), F_{\mathrm{edge}}(G)\right) \\ &= 1 - \mathrm{IoU}'\left(F_{\mathrm{edge}}(P), F_{\mathrm{edge}}(G)\right) \\ &= 1 - \text{B-IoU}'(P, G). \end{aligned} \tag{14}$$

Here, for optimizing $\theta(S_1)$ in the B3SM, the Canny edge detection filter with a monotone gradient condition, a minimum threshold of 0.5, a maximum threshold of 1.0, and an aperture size of 3 is utilized to the rectified predicted segmentation map of which values of elements are in [0, 1] [37], [38]. In addition, the dilation and erosion filters with a size of 3 × 3 and a value of 3 are utilized to calculate F_{edge}. Note that F_{edge} is differentiable in this case because the convolutional filters of F_e, F_{canny}, and F_d are differentiable.

Since B-Loss is more sensitive than IoU-Loss for the precise segmentation of boundaries of buildings, the more accurate segmentation of buildings with clearer boundaries can be achieved using B-Loss in B3SM. B-IoU is a more pessimistic evaluation metric than other boundary-oriented evaluation metrics of BF and BJ, and the area-oriented evaluation metric of IoU. Fig. 3 illustrates the examples of the noise added segmentation and boundary maps. The images at Noise = 0 are the ground truth images. By adding sharp areas in the boundaries, segmentation maps that are only mispredicted in the boundaries but still exhibit high IoU values are generated. As shown in Fig. 4, the boundary-oriented evaluation metrics are more sensitive to the noisy boundaries than the IoU. Furthermore, among the boundary-oriented evaluation metrics, the B-IoU is most sensitive to the noisy boundaries compared to any other

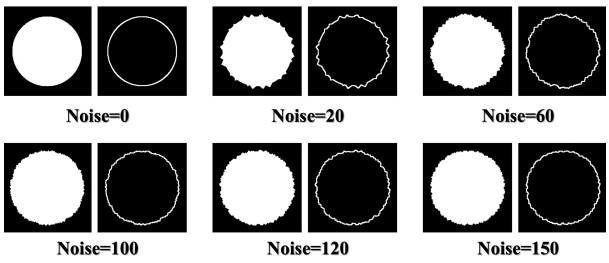


Fig. 3. Sample images of the noise-added segmentation and boundary maps.

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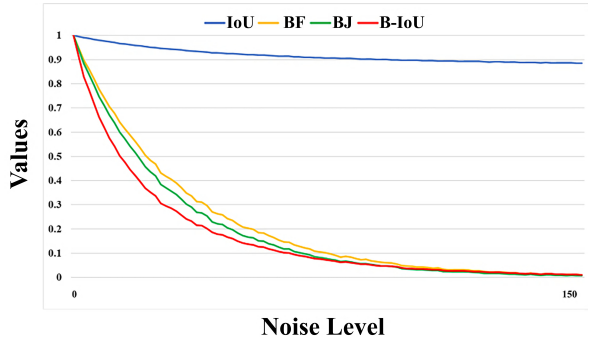


Fig. 4. Values of IoU, BF, BJ, and B-IoU along with a noise level at the boundaries of segmentation maps.

evaluation metrics. Here, BF and BJ exhibit relatively higher values than B-IoU when the shapes of predicted buildings become worse due to the noise. That is, BF and BJ are less sensitive than B-IoU. In contrast, B-IoU remarkably decreases along with the noise level at the boundaries of buildings and exhibits a lower value than other evaluation metrics. Therefore, B-IoU is a more pessimistic and sensitive evaluation metric for effectively assessing the accuracy of boundaries of segmented buildings than other metrics. The mathematical proof is demonstrated in the Supplementary Material.

4) Implementation of Loss Functions Into B3SM: In the B3SM, BCE-Loss and B-Loss are used to optimize S_1 to produce prepredicted segmentation maps with enhanced boundaries. B-Loss is implemented in S_1 to extract boundary-oriented features and then generate enhanced shapes for buildings. However, B-Loss alone is not suited for fully optimizing deep learning architecture. It was found that the B-Loss results in the generation of several misclassified empty regions in segmented buildings. Therefore, BCE-Loss is implemented in combination with B-Loss to compensate for these misclassified empty regions in the buildings. Meanwhile, IoU-Loss is used in $S_{1,2}$ to train the B3SM to improve the recognition of segmented building shapes from aerial images. In S_1, BCE-Loss, B-Loss, and IoU-Loss are used for supplementary training to extract the boundaries of buildings from aerial images. The optimization of S_2 using IoU-Loss enables better prediction of building shapes based on the features extracted from building boundaries in S_1. Therefore, the B3SM is optimized as follows:

$$\begin{aligned} S_1&: \min_{\theta(S_1)} \left[L_{\text{B-Loss}}\left(P_{S_1}, G; \theta(S_1)\right) + L_{\text{BCE}}\left(P_{S_1}, G; \theta(S_1)\right)\right] \\ S_2&: \min_{\theta(S_1 \cup S_2)} \left[L_{\text{IoU}}\left(P_{S_2}, G; \theta(S_1 \cup S_2)\right)\right]. \end{aligned} \tag{15}$$

When the B3SM is fully optimized with the parameter sets of $\Theta(S_1)$ and $\Theta(S_2)$, it generates a final predicted segmentation map as follows:

$$\begin{aligned} P_{S_1}^{(H,W,1)} &= f_{S_1}\left(I^{(H,W,3)}; \Theta(S_1)\right) \\ P_{S_2}^{(H,W,1)} &= f_{S_2}\left(\mathcal{M}\left(I^{(H,W,3)}, P_{S_1}^{(H,W,1)}\right); \Theta(S_2)\right) \\ &\cong G^{(H,W,1)}. \end{aligned} \tag{16}$$

IV. Data Source

The Inria public dataset is considered in this study [17]. The Inria dataset covers an area of 405 km^2. The

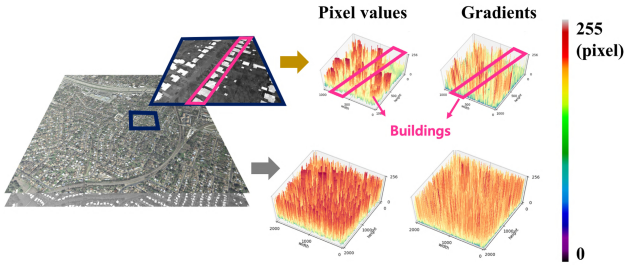


Fig. 5. Aerial image and gradient analysis. The pink color indicates the same building areas on left and right. The pixel values and gradients of an aerial image are nonuniformly distributed, regardless of the buildings and background. The buildings in the box have different pixel values.

RGB images in the Inria dataset have a spatial resolution of 0.3 m and cover five cities (Austin, Chicago, Kitsap, Tyrol, and Vienna). We randomly cropped the images to a size of 256 × 256 pixels from the original size of 5000 × 5000 pixels, and 28 800 images were considered for each category, resulting in a total of 144 000 images. Dataset was divided into training, validation, and test sets for k-fold cross validation [39]. Specifically, the images of three cities were used for the training set, the images of another city were used for the validation set, and the images of the other city were used for the test set. Therefore, the numbers of images in the training, validation, and test sets were 86 400, 28 800, and 28 800, respectively.

We also trained deep learning architectures for semantic segmentation using 9000 images from domestic locations in Seoul, Suwon, and Daegu, which are the most complex cities in South Korea. This dataset consists of RGB aerial images with a spatial resolution of 0.61 m and the corresponding ground-truth labels for binary classes (background and building). These images have relatively low spatial resolutions compared to those used in previous studies (0.51 m [14] and 0.30 m [17]). The dataset covers an area of 147.46 km^2 and is divided into an area of 110.59 km^2 for training and an area of 36.87 km^2 for testing. The numbers of images in the training, validation, and test sets are 4500, 2250, and 2250, respectively.

Fig. 5 presents an aerial image and its corresponding gradient analysis. One can see that the shapes and colors of the buildings vary, meaning that the pixel intensities of the image are nonuniformly distributed regardless of the localization of buildings. These characteristics may degrade the performance of threshold-based conventional computer vision algorithms for segmenting buildings in aerial images. Furthermore, because the gradients [40], which are the indications of the strength and the direction of the slope at each point and are used for illustrating robust features of images, are disproportionately distributed irrespective of the boundaries, segmenting buildings with accurate boundaries and shapes requires advanced algorithms for image analysis beyond conventional computer vision algorithms.

V. Experimental Results

In this section, we carry out the following experiments to evaluate the performance of B3SM: 1) two-scheme learning

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TABLE II

EVALUATION METRICS AND THEIR DEFINITIONS. P AND G INDICATE THE BINARIZED IMAGES OF PREDICTION AND GROUND TRUTH, RESPECTIVELY

Evaluation Metric	Definitions
Intersection over Union (IoU) [17]	$\text{IoU}(P, G) = \frac{G \cap P}{G \cup P}$
Boundary-oriented IoU (B-IoU)	$\text{B-IoU}(P, G) = \text{IoU}(F_{\text{edge}}(P), F_{\text{edge}}(G)) = \frac{F_{\text{edge}}(G) \cap F_{\text{edge}}(P)}{F_{\text{edge}}(G) \cup F_{\text{edge}}(P)}$

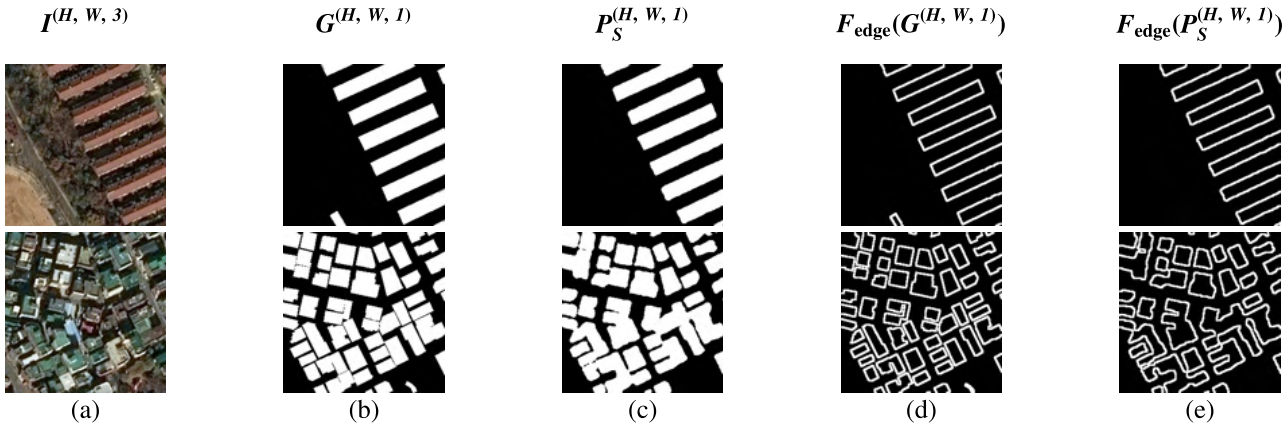


Fig. 6. Example images for evaluation of IoU and B-IoU values. (a) Input image, (b) ground truth, (c) prediction, (d) edge-detected ground truth, and (e) edge-detected prediction.

is compared to one-scheme learning; 2) the performance of deep-learning models with B-Loss and other loss functions is compared; 3) the USIM and other upsampling operations are compared; and 4) the performance of B3SM and other state-of-the-art models for building segmentation in aerial images is compared in terms of IoU and B-IoU, and the associated ablation study is conducted.

A. Experimental Setup

In this experiment, we trained and tested the proposed model with two datasets, which are illustrated in Section IV. A k-fold cross-validation method was applied to both datasets. A total of 144 000 images in Dataset I, i.e., the Inria dataset [17], were divided into 86 400 for the training set, 28 800 for the validation set, and 28 800 for the test set. A total of 9000 images in Dataset II, i.e., our dataset, were divided into 4500 for the training set, 2250 for the validation set, and 2250 for the test set. Our experiments were conducted using the public platform Tensorflow [41]. The implemented deep learning architectures were trained using an Adam optimizer [42] with an SGD [43] method and a minibatch of 12. For the parameters of the Adam optimizer, a learning rate was initially set to 1e-3 and then divided by 2 in every 50 epochs. In addition, decay rates (β_1 and β_2) were set to 0.9 and 0.999, respectively, and ϵ was 1e-8. Group normalization was applied instead of batch normalization because the batch size is too small in terms of memory utilization [44], [45]. The G of Group normalization was 32, and the initial values of group normalization were set to exhibit Gaussian distribution. Every deep learning model was trained with 200 epochs. The Tensorflow server consisted of two-way CPUs, 32 GB of RAM, and four-way Titan XP (11 GB) GPUs.

B. Evaluation Metric

The IoU and B-IoU metrics were utilized to evaluate the performance of our model. B-IoU is used to assess the shapes of predicted buildings because IoU is not always suitable for the quantification of shapes. B-IoU is the IoU value between the boundary-detected ground truth $[F_{\text{edge}}(G^{(H,W,1)})]$ and the boundary-detected prediction $[F_{\text{edge}}(P_S^{(H,W,1)})]$.

Therefore, it represents the intersection area between two boundaries. A dilation method was applied to the edge-detected images of predicted maps because the thicknesses of the boundaries of the buildings were too small to measure the intersection area between the ground truth and the prediction.

Fig. 6 presents the evaluation results for segmented images in terms of IoU and B-IoU. In Fig. 6, the pixel accuracies of the predicted segmentation maps are 98.94% (top) and 96.04% (bottom), and the IoU values of the maps are 95.99% (top) and 92.73% (bottom). The values for the predicted results are very high for both images. However, the shapes of the predicted images in Fig. 6(c) and (e) (bottom) are significantly different from those in the corresponding ground-truth results in Fig. 6(b) and (d), respectively. The predicted segmentation map for the top image yields a B-IoU value of 87.47%, whereas the predicted segmentation map for the bottom image yields a B-IoU value of 70.07%. As expected, the B-IoU values of the images are significantly different. These results demonstrate that the B-IoU values of predicted segmentation maps better represent whether the shapes of

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TABLE III

QUANTITATIVE COMPARISON OF THE IOU AND B-IOU VALUES OF TWO-SCHEME LEARNING AND ONE-SCHEME LEARNING WITH DIFFERENT LOSS FUNCTIONS. BCE-LOSS AND IOU-LOSS ARE UTILIZED FOR LOSS-A AND LOSS-B, RESPECTIVELY. LOSS-C ACCOMMODATES BCE-LOSS FOR S_1 AND IOU-LOSS FOR S_2, AND LOSS-D UTILIZES BCE-LOSS AND B-LOSS FOR S_1 AND IOU-LOSS FOR S_2. THE HIGHEST VALUE AND IMPROVEMENT ARE MARKED AS **BOLD**. THE VALUES IN SMALL BRACKETS INDICATE THE IMPROVEMENTS FROM LOSS-B

Evaluation Metrics	Model	One- scheme Learning		Two- scheme Learning	
		Loss-A	Loss-B	Loss-C	Loss-D
		BCE-Loss	IoU-Loss	S_1: BCE-Loss S_2: IoU-Loss	S_1: BCE-Loss + B-Loss S_2: IoU-Loss
IoU	FCN	62.18	62.71	65.25 (+2.54)	62.89 (+0.18)
	SegNet	63.00	63.21	67.83 (+4.62)	67.18 (+3.97)
	U-Net	62.96	64.07	69.85 (**+4.78**)	68.39 (**+4.32**)
	FusionNet	65.62	67.27	**71.13** (+3.86)	**69.55** (+2.28)
B-IoU	FCN	63.93	63.20	64.56 (+0.62)	67.88 (+3.94)
	SegNet	64.21	64.29	65.11 (+0.82)	69.02 (**+4.73**)
	U-Net	64.19	64.83	65.72 (**+0.89**)	68.65 (+3.82)
	FusionNet	64.35	66.73	**66.78** (+0.05)	**70.87** (+4.14)

segmented buildings are similar to those in the corresponding ground-truth results than the IoU values. It should be noted that the ultimate goal of our proposed deep learning model is to predict better shapes of segmented buildings in aerial images.

Furthermore, we compared B-IoU with other boundary-oriented state-of-the-art evaluation metrics, such as BJ [33] and a boundary F1 (BF) score [46]. Since there is no assessment method to quantify the performance of evaluation metrics, we investigated which evaluation metrics are capable of offering the most useful information as a boundary-oriented evaluation metric score. As a result, it was found that the B-IoU is a more suitable boundary-oriented evaluation metric than BJ and BF. A detailed explanation is shown in the Supplementary Material.

C. Experimental Results

To compare the performance of our proposed deep learning architecture to that of other state-of-the-art models in building segmentation tasks, quantitatively, we compute the IoU and B-IoU values for each model and plot a precision–recall (PR) curve. Also, the performances of subparts of B3SM—two-scheme learning, B-Loss, and 3) USIM—are evaluated to validate the novelty of the subparts in a building segmentation task.

1) Two-Scheme Learning: To evaluate the performance of two-scheme learning for building segmentation tasks, two-scheme learning was compared to one-scheme learning with different baseline deep learning models and loss functions. To evaluate the individual performance of two-scheme learning, the USIM was removed, and the same baseline architectures were reused. Table III lists the IoU and B-IoU values of the prediction results for one-scheme learning and two-scheme learning, respectively. The deep learning architectures in Table III are implemented for both one-/two-scheme learning to evaluate the performance of two-scheme learning. For one-scheme learning, deep learning models (D_m) listed in Table III are implemented as follows:

$$P_{S_1}^{(H,W,1)} = D_m\left(I^{(H,W,3)}; \theta(S_1)\right) \tag{17}$$

and the deep learning models are optimized by the loss function illustrated in Table III using the prepredicted segmentation map ($P_{S_1}^{(H,W,1)}$) and the ground truth ($G^{(H,W,1)}$), respectively. On the contrary, in two-scheme learning, the deep learning model is repeatedly implemented as follows:

$$P_{S_2}^{(H,W,1)} = D_m(\mathcal{M}\left(D_m\left(I^{(H,W,3)}; \theta(S_1)\right); \theta(S_2)\right) \tag{18}$$

and the deep learning model is then optimized by the generated segmentation map ($P_{S_2}^{(H,W,1)}$) and $G^{(H,W,1)}$. In Table III, Loss-A indicates that BCE-Loss is utilized in one-scheme learning, whereas Loss-B indicates that IoU-Loss is utilized in one-scheme learning. For two-scheme learning, the shape-oriented Loss-C represents BCE-Loss in S_1 and IoU-Loss in S_2 for segmenting the areas of buildings during training. Loss-C is defined as follows:

$$\begin{aligned} S_1: &\ \underset{\theta(S_1)}{\text{minimize}} \left[\mathcal{L}_{\text{BCE}}\left(f_{S_1}(I), G; \theta(S_1)\right)\right] \\ S_2: &\ \underset{\theta(S_2)}{\text{minimize}} \left[\mathcal{L}_{\text{IoU-Loss}}\left(f_{S_2}\left(P_{S_1}\right), G; \theta(S_2)\right)\right]. \end{aligned} \tag{19}$$

Similarly, the boundary-oriented Loss-D utilizes both B-Loss and BCE-Loss in S_1 and IoU-Loss in S_2 to derive clear boundaries for buildings. Loss-D is defined as follows:

$$\begin{aligned} S_1: &\ \underset{\theta(S_1)}{\text{minimize}} \left[\mathcal{L}_{\text{B-Loss}}\left(f_{S_1}(I), G; \theta(S_1)\right)\right. \\ &\qquad \left. + \mathcal{L}_{\text{BCE}}\left(f_{S_1}(I), G; \theta(S_1)\right)\right] \\ S_2: &\ \underset{\theta(S_2)}{\text{minimize}} \left[\mathcal{L}_{\text{IoU-Loss}}\left(f_{S_2}\left(P_{S_1}\right), G; \theta(S_2)\right)\right]. \end{aligned} \tag{20}$$

As summarized in Table III, the greatest IoU value is achieved when FusionNet [47] is trained using Loss-C.

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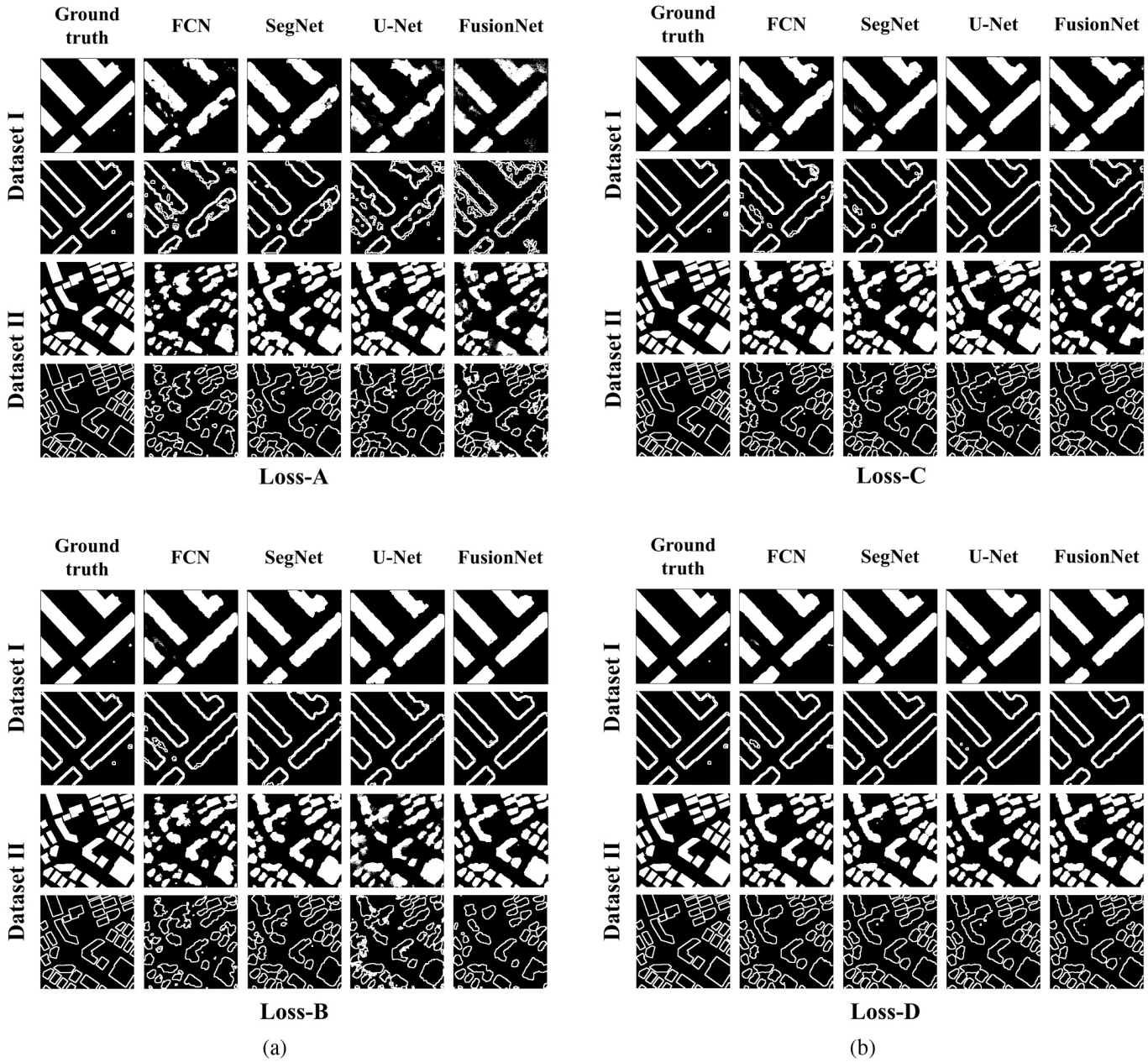


Fig. 7. Segmentation results obtained by one-scheme learning and two-scheme learning with different models and loss functions. (a) One-scheme learning with Loss-A (the first row) and Loss-B (the second row). (b) Two-scheme learning with Loss-C (the first row) and Loss-D (the second row). FCN (the first column), SegNet (the second column), U-Net (the third column), and FusionNet (the fourth column) are utilized as baseline architectures, respectively.

The IoU-value is 71.13%, which is 3.86% greater than that of one-scheme learning with Loss-B. The images predicted by two-scheme learning with Loss-C in the first row of Fig. 7(b) exhibit improved segmentation performance for Datasets I and II. In addition, Table III reveals that Loss-D results in an improved B-IoU value. SegNet [48], which is constructed using two-scheme learning with Loss-D, significantly improves the B-IoU value to 69.02%, which is 4.73% greater than that of one-scheme learning with Loss-B. The predicted images from the two-scheme learning models with Loss-D [see Fig. 7(b)] exhibit more precise boundaries compared to the buildings segmented using one-scheme learning for Datasets I and II.

For quantitative performance analysis of two-scheme learning, the PR curves of the IoU and B-IoU values of the segmented buildings are plotted in Fig. 8. Fig. 8(a)–(d) presents the PR curves of IoU values, and Fig. 8(e)–(h) presents the PR curves of B-IoU values. As shown in Fig. 8, the models with Loss-C segment buildings precisely, whereas the models with Loss-D generate clear boundaries for buildings in segmented images. These results demonstrate that two-scheme learning is more suitable for segmenting buildings in aerial images compared to one-scheme learning.

2) B-Loss: To evaluate the effect of B-Loss on the training of scheme, the B-IoU values of U-Net and SegNet optimized loss functions with and without B-loss are compared.

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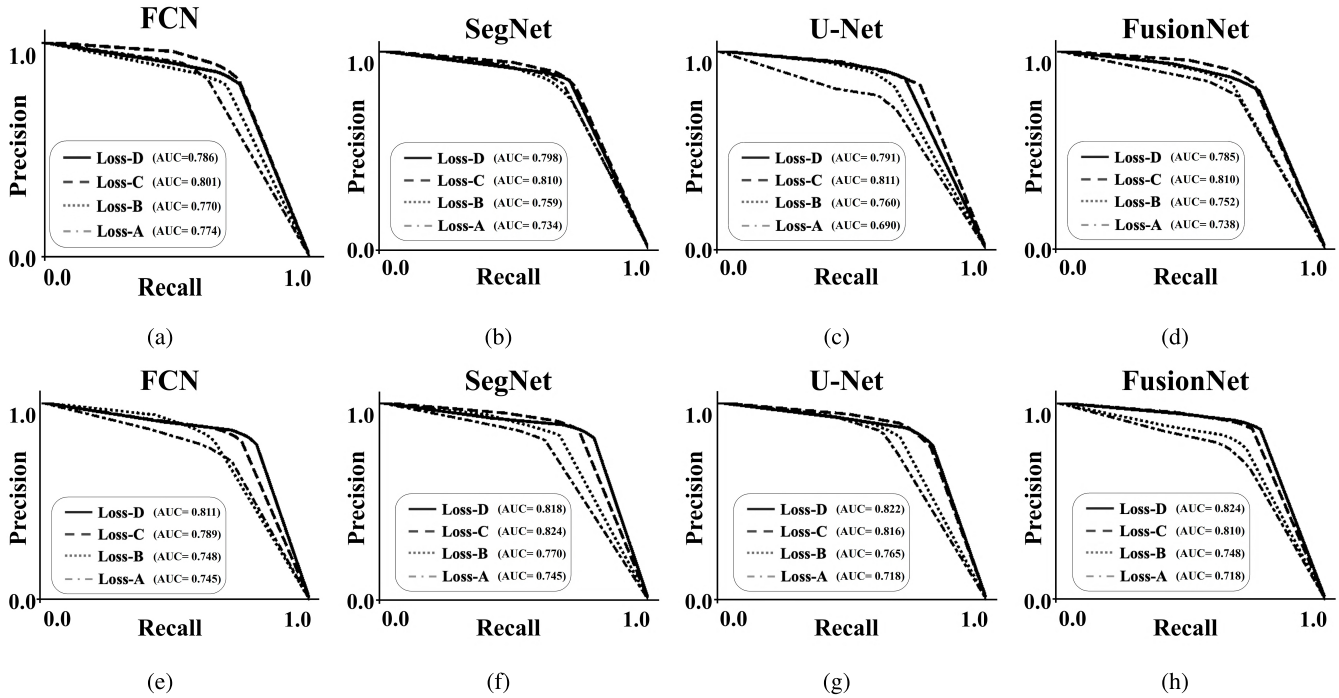


Fig. 8. PR curves of two-scheme learning with different loss functions and baseline models for semantic segmentation and boundary shape prediction. (a)–(d) Semantic segmentation (IoU). (e) and (f) Boundary shape of buildings (B-IoU). PR-AUCs are in the legends with loss functions.

TABLE IV

PREDICTED RESULTS FROM U-NET AND SEGNET CONSTRUCTED BY DIFFERENT LOSS FUNCTIONS WITH AND WITHOUT B-LOSS. IN THE CASE OF THE BASELINE, THE DEFAULT LOSS FUNCTIONS ARE UTILIZED. IN THE CASE OF B-LOSS ADDITION, THE B-LOSS IS UTILIZED WITH THE BASELINE LOSS FUNCTIONS

			L_2-Loss	BCE-Loss	IoU-Loss
U-Net	Baseline	IoU	62.58	62.96	64.07
		BIoU	63.89	62.96	64.83
	B-Loss Addition	IoU	62.70 (+0.12)	**63.50 (+0.54)**	64.11 (+0.04)
		BIoU	65.92 (+2.03)	66.23 (+2.04)	**67.17 (+2.34)**
SegNet	Baseline	IoU	61.17	63.00	63.21
		BIoU	64.33	64.21	64.29
	B-Loss Addition	IoU	61.87 (**+0.70**)	63.01 (+0.01)	**63.86** (+0.65)
		BIoU	65.78 (+1.45)	66.10 (+1.89)	**67.02 (+2.73)**

Fig. 9 illustrates the predicted results. Fig. 9(a) illustrates the segmented maps and their corresponding boundary-detected maps obtained using U-Net with baseline loss functions, whereas Fig. 9(b) illustrates the segmented maps and their corresponding boundary-detected maps obtained using U-Net with the combination of B-Loss and the baseline loss functions. Fig. 9(c) illustrates the segmented maps and their corresponding boundary-detected maps obtained using SegNet with baseline loss functions, whereas Fig. 9(d) illustrates the segmented maps and their corresponding boundary detected maps obtained using SegNet with B-Loss. As shown in Fig. 9, the deep learning architectures with B-Loss exhibit more accurate boundaries of buildings in aerial images than those without B-loss.

Table IV illustrates quantitative results for the comparison of the performance of B-Loss and other loss functions. In the experiments, U-Net and SegNet, which are optimized by B-Loss with baseline loss functions, offered the higher IoU and B-IoU values than U-Net and SegNet, optimized by the only baseline loss functions. Those results demonstrate that the B-Loss can improve the performance of deep learning models for segmenting boundaries of buildings.

3) Comparisons of USIM and Other Interpolation Operations: The USIM was compared to other upsampling operations, namely, nearest-neighbor interpolation and bicubic interpolation, to investigate advantages of the USIM over these methods. Because conventional interpolation methods cannot combine two images into a single image but only double the sizes of images, overlaying operations should be applied to interpolation operations. Nearest-neighbor interpolation with an overlay operation is denoted as NNO. Bicubic interpolation with an overlay operation is denoted as BCO.

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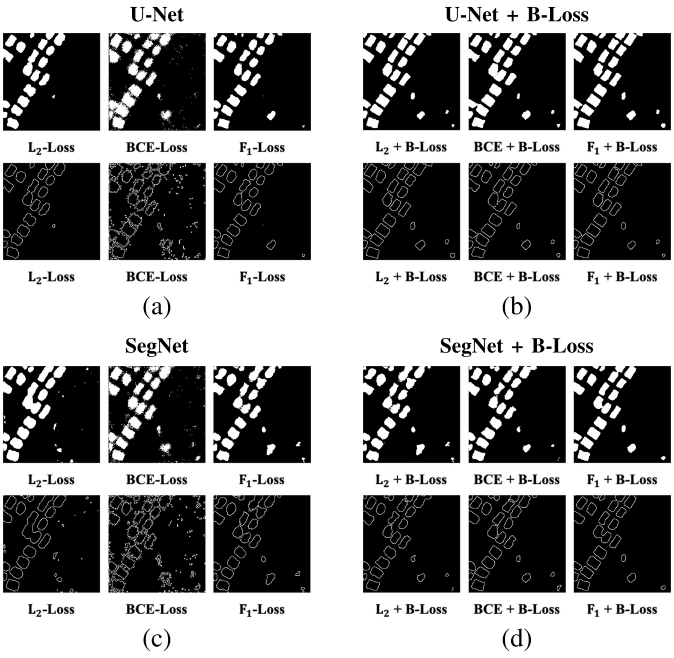


Fig. 9. Segmentation results of one-scheme learning with different loss functions and baseline architectures. (a) and (b) are obtained using U-Net as the baseline architecture. (c) and (d) are obtained using SegNet as the baseline architecture.

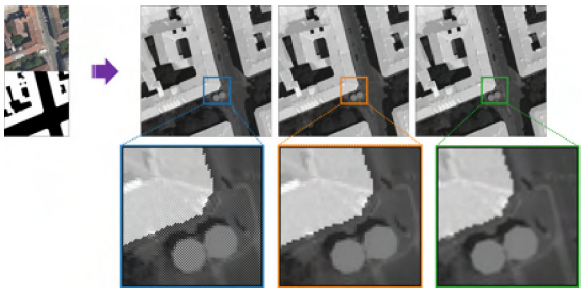


Fig. 10. Output images of (Left) USIM, (Middle) nearest-neighbor interpolation, and (Right) bicubic interpolation operations. The blue solid rectangle indicates the output image of USIM. The orange and green solid rectangles indicate the output images of nearest-neighbor interpolation and bicubic interpolation, respectively.

Fig. 10 presents output images from the USIM, NNO, and BCO. In the magnified images, one can see that the images obtained from the USIM preserve the original pixel values of $I^{(H,W,3)}$ and $G^{(H,W,1)}$, whereas the images obtained using the other operations lose the original pixel values of the $I^{(H,W,3)}$ and $G^{(H,W,1)}$. The distributions of pixel values for images merged by the USIM, NNO, and BCO are illustrated in the Supplementary Material.

To analyze the advantages of the USIM quantitatively, the pixel accuracy (PA), the pixel distance rate (PDr), and the pixel absolute distance rate (PADr) are examined along with pixel contrast since aerial images have various pixel contrasts depending on regions and times. Fig. 11 illustrates the evaluation scores along with the pixel contrast of images when the merging operation ($\mathcal{M}$) is USIM, NNO, or BCO. PA is utilized to evaluate the similarity between the pixel values of two images. When ${I_1}^{(H,W,1)}$ and ${I_2}^{(H,W,1)}$ are the same, the value of PA is one. In contrast, the PA value is zero when ${I_1}^{(H,W,1)}$ and ${I_2}^{(H,W,1)}$ are completely different. As shown in Fig. 11(a), the PA value for the USIM is 0.5, whereas the PA values for the other interpolation methods are less than 0.04. Because the USIM preserves the pixel values of each ${I_1}^{(H,W,1)}$ and ${I_2}^{(H,W,1)}$, the PA values of the USIM are enhanced significantly. Similarly, PDr and PADr values can also be utilized to measure the similarity between two images. As shown in Fig. 11(b) and (c), the images merged by the USIM exhibit lower PDr and PADr values than those merged by the other compared methods. The low PDr and PADr values for the USIM indicate that it preserves the original features of input images in output images. It should be noted that the PDr and PADr values decrease when two images are more similar.

TABLE V

SSIM, MEAN SQUARED ERROR (MSE), AND PA BETWEEN IMAGES GENERATED BY DIFFERENT UPSAMPLING AND MERGING OPERATIONS

	SSIM	MSE	PA
USIM	**0.383**	927.9	**0.500**
Nearest-Neighbor	0.369	931.288	0.034
Bi-cubic	0.381	**927.8**	0.035

Furthermore, the similarities between original ($I^{(H,W,3)}$ and $G^{(H,W,1)}$) and merged images $[\mathcal{M}(I^{(H,W,3)}, G^{(H,W,1)})]$ obtained using the USIM, NNO, and BCO were also evaluated. As summarized in Table V, the USIM results in the highest structural similarity (SSIM) (0.383) [49] and PA (0.5) values compared to other operations. These results suggest that the output images of the USIM have pixel values that are similar to those of the input images ${I_2}^{(H,W,1)}$ and $G^{(H,W,1)}$. Here, the higher SSIM and PA of USIM suggest that USIM allows better effective feature extraction than other operations while preserving rich color and spatial information on original input images [50]–[52].

Another advantage of the USIM is that the features extracted from the boundaries of buildings are well preserved, as mentioned previously. For quantitative analysis, the receiver operating characteristic (ROC) curves of the USIM, NNO, and BCO were plotted during the extraction of features from building boundaries. Fig. 12(a) presents the ROC curves of the USIM, NNO, and BCO. The AUC values of the USIM, NNO, and BCO are 0.833, 0.783, and 0.775 for the public dataset. Similarly, the AUC values of the USIM, NNO, and BCO are 0.897, 0.876, and 0.598, respectively, for our custom dataset, as shown in Fig. 12(b). These results reveal that the USIM more accurately preserves the boundaries of buildings in merged images compared to NNO and BCO.

Fig. 12(c) and (d) presents the edge-detected images of the boundaries of buildings obtained by F_{edge} when the USIM, NNO, and BCO are applied to input images. The edge-detected images are generated by the following equation:

$$I_{\text{edge}} = F_{\text{edge}}\left(\mathcal{M}\left(I^{(H,W,3)}, G^{(H,W,1)}\right)\right) \tag{21}$$

where I_{edge} is an edge-detected image of an image merged by a given operation, ($\mathcal{M}$). In F_{edge}, a simple Canny edge

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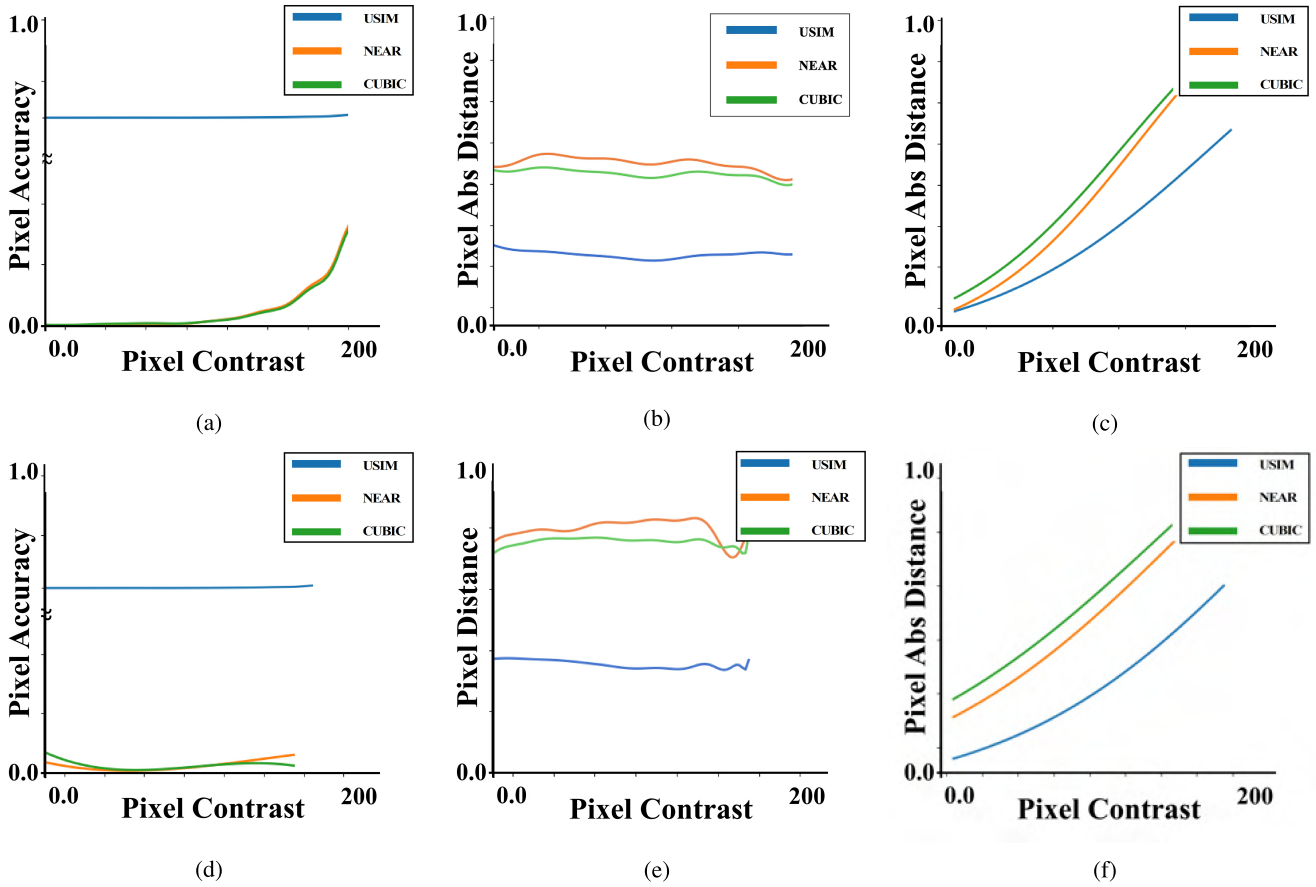


Fig. 11. Quantitative comparisons of USIM and other interpolation methods. (a)–(c) Analytic graphs of PA, PDr, and PADr for the merging operations on Inria dataset. (d)–(f) Analytic graphs of PA, PDr, and PADr for the merging operations on our dataset.

TABLE VI

EVALUATION METRICS OF THE PROPOSED B3SM COMPARED TO OTHER STATE-OF-THE-ART MODELS, INCLUDING ADAPTED-FCN, U-NETPPL, MC-FCN, AND MS-CDN

Network	Precision		Recall		OA		F1-score		IoU		BIoU	
	DI	DII	DI	DII	DI	DII	DI	DII	DI	DII	DI	DII
Adapted-FCN	81.09	81.71	79.02	78.98	97.89	97.93	80.04	80.32	66.72	67.12	67.42	69.74
U-NetPPL	82.35	82.36	80.49	84.89	98.07	98.33	81.41	83.60	68.65	71.82	69.92	72.35
MC-FCN	82.16	82.16	80.68	83.77	98.08	98.27	81.41	82.95	68.66	70.87	71.01	73.55
MS-CDN	83.21	83.22	81.71	85.05	98.23	98.42	82.45	84.13	70.14	72.60	71.22	73.95
B3SM (Ours)	**85.24**	**86.08**	**82.68**	**86.76**	**98.38**	**98.74**	**83.94**	**86.42**	**72.32**	**76.08**	**75.22**	**78.16**
Max. Improvement	+4.15	+4.37	+3.66	+7.77	+0.49	+0.80	+3.90	+6.09	+5.60	**+8.97**	+7.80	**+8.42**

detection filter is utilized to find the edges in an image. The USIM yields more accurate building edges compared to the other methods. These results suggest that the USIM provides a better feature extraction from building boundaries compared to other merging operations. Overall, the USIM preserves the original pixel values of input images accurately and highlights the features of the boundaries of buildings in output images.

4) B3SM: The proposed B3SM was applied to segment buildings in complex aerial images, and the results were compared to those of other state-of-the-art models, namely, the Adapted-FCN [23], U-NetPPL [14], MC-FCN [2], and MS-CDN [16] models. To compare the B3SM to the other models on building segmentation tasks quantitatively, we computed IoU and B-IoU values and plotted the PR curves for the B3SM and the other state-of-the-art models (see Fig. 13). In the B3SM, the baseline architecture of S_1 was FusionNet, whereas a modified FusionNet [47], in which the residual connections were replaced with the USIM, was utilized as the baseline architecture of S_2. The loss functions of S_1 were BCE-Loss and B-Loss, while IoU-Loss was utilized as the loss function of S_2.

Table VI summarizes the quantitative analysis results for the B3SM, Adapted-FCN, U-NetPPL, MC-FCN, and MS-CDN. The B3SM improves the IoU values by 5.60% and 8.97% for Datasets I and II, respectively. Similarly, the B-IoU values of B3SM are improved by 7.8% and 8.42% for Datasets I and II, respectively, compared to those of the other state-of-the-art models. The improved B-IoU and IoU values of the B3SM

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TABLE VII

RESULTS OF THE ABLATION STUDY OF B3SM THAT HAS THREE MAIN COMPONENTS: TWO-SCHEME LEARNING, USIM, AND B-LOSS. THE BEST PERFORMANCE IS IN BOLD AND THE SECOND ONE IN UNDERLINE

Description	**Dataset I**		**Dataset II**	
Scheme + merge operation + (loss functions S_1) + (loss functions S_2)	**IoU**	**BIoU**	**IoU**	**BIoU**
Two-Scheme + USIM + (BCE-Loss + B-Loss) + (IoU-Loss + B-Loss)	<u>72.17</u>	**75.22**	<u>75.98</u>	**78.20**
Two-Scheme + USIM + (BCE-Loss + B-Loss) + (IoU-Loss)	**72.32**	**75.22**	**76.08**	<u>78.16</u>
Two-Scheme + USIM + (BCE-Loss) + (IoU-Loss + B-Loss)	71.29	74.21	75.70	78.04
Two-Scheme + USIM + (BCE-Loss) + (IoU-Loss)	72.18	72.78	76.01	77.15
Two-Scheme + Concate + (BCE-Loss + B-Loss) + (IoU-Loss + B-Loss)	70.27	74.73	75.70	77.94
Two-Scheme + Concate + (BCE-Loss + B-Loss) + (IoU-Loss)	70.84	73.14	75.17	77.53
Two-Scheme + Concate + (BCE-Loss) + (IoU-Loss + B-Loss)	69.92	<u>74.82</u>	74.63	77.67
Two-Scheme + Concate + (BCE-Loss) + (IoU-Loss)	71.48	73.71	75.81	77.02
Two-Scheme + Overlay + (BCE-Loss + B-Loss) + (IoU-Loss + B-Loss)	69.25	70.15	72.26	74.55
Two-Scheme + Overlay + (BCE-Loss + B-Loss) + (IoU-Loss)	70.12	70.97	72.47	73.67
Two-Scheme + Overlay + (BCE-Loss) + (IoU-Loss + B-Loss)	70.77	71.24	72.16	74.29
Two-Scheme + Overlay + (BCE-Loss) + (IoU-Loss)	70.82	70.91	72.85	73.16
Scheme + Upscale Operation + Loss functions	**IoU**	**BIoU**	**IoU**	**BIoU**
One-Scheme + USIM + (IoU-Loss + B-Loss)	71.91	73.23	74.28	76.37
One-Scheme + USIM + (BCE-Loss + B-Loss)	70.63	73.91	73.95	75.81
One-Scheme + USIM + (IoU-Loss)	71.63	72.17	73.16	75.12
One-Scheme + USIM + (BCE-Loss)	70.93	72.53	74.10	74.93
One-Scheme + Deconvolution + (IoU-Loss + B-Loss)	71.29	71.87	73.94	74.65
One-Scheme + Deconvolution + (BCE-Loss + B-Loss)	69.75	71.91	73.92	74.83
One-Scheme + Deconvolution + (IoU-Loss)	69.83	71.22	73.02	74.93
One-Scheme + Deconvolution + (BCE-Loss)	69.37	71.51	73.17	74.29

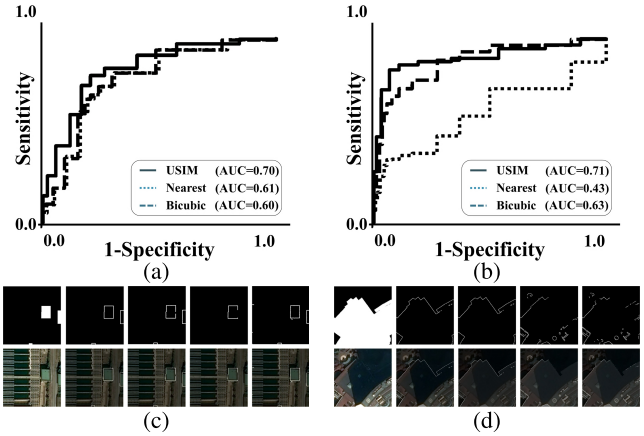


Fig. 12. ROC curves and edge-detected images for the USIM, NNO, and BCO for feature extraction from building boundaries: ROC curve of the USIM, NNO, and BCO for (a) Dataset I and (b) Dataset II. Edge-detected images using (c) Dataset I and (d) Dataset II of building boundaries in output images from the USIM, NNO, and BCO. The ground-truth and raw input images are presented in the first column. The second column shows the edge-detected image from (Top) ground truth and (Bottom) overlay of the raw input image and the edge-detected image, whereas the third, fourth, and fifth columns show the edge-detected images of output images from (Top) USIM, NNO, and BCO and (Bottom) overlay of the raw input images and the edge-detected images, respectively.

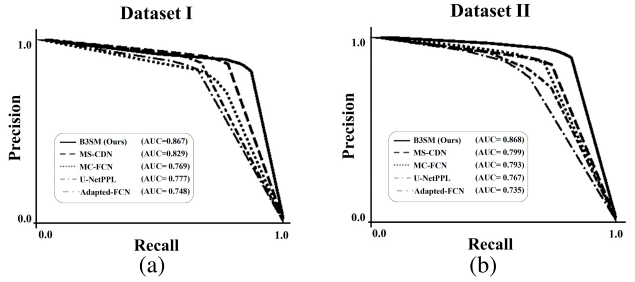


Fig. 13. PR curves of the proposed B3SM and other state-of-the-art models for the boundary shape prediction of buildings. AUC values are presented in the legends for each deep learning model. (a) PR curve for Dataset I. (b) PR curve for Dataset II.

indicate that the B3SM predicts the boundaries and shapes of buildings in aerial images more accurately.

In addition, we calculated the AUC values of the B3SM, Adapted-FCN, U-NetPPL, MC-FCN, and MS-CDN (Fig. 13). For Dataset I, the AUC value of the B3SM is 0.867, whereas the values of MS-CDN, MC-FCN, U-NetPPL, and Adapted-FCN are 0.829, 0.769, 0.777, and 0.748, respectively [Fig. 13(a)]. Similarly, for Dataset II, the AUC value of the B3SM is 0.868, whereas the values of MS-CDN, MC-FCN, U-NetPPL, and Adapted-FCN are 0.799, 0.793, 0.767, and 0.735, respectively [Fig. 13(b)].

In order to evaluate the effectiveness of proposed components in B3SM, we performed an ablation study for the two-scheme learning, USIM, and B-Loss. Table VII illustrates the results of the ablation study for B3SM. As mentioned previously, FusionNet is utilized as the baseline architecture. In two-scheme learning, the basic architecture of FusionNet is utilized as the first baseline architecture in S_1, and the modified FusionNet of which all deconvolution operations are replaced by USIM is utilized in the second architecture of S_2. On the contrary, in one-scheme learning, FusionNet with the illustrated upscale operation is utilized. All deconvolution operations in the decoder part of FusionNet are replaced by USIM and other upscale operations for one-scheme learning. The ablation study demonstrates that the performance of B3SM is improved by three main components; two-scheme learning, USIM, and B-Loss. Two-scheme learning improved IoU and B-IoU values in most cases, especially when it is

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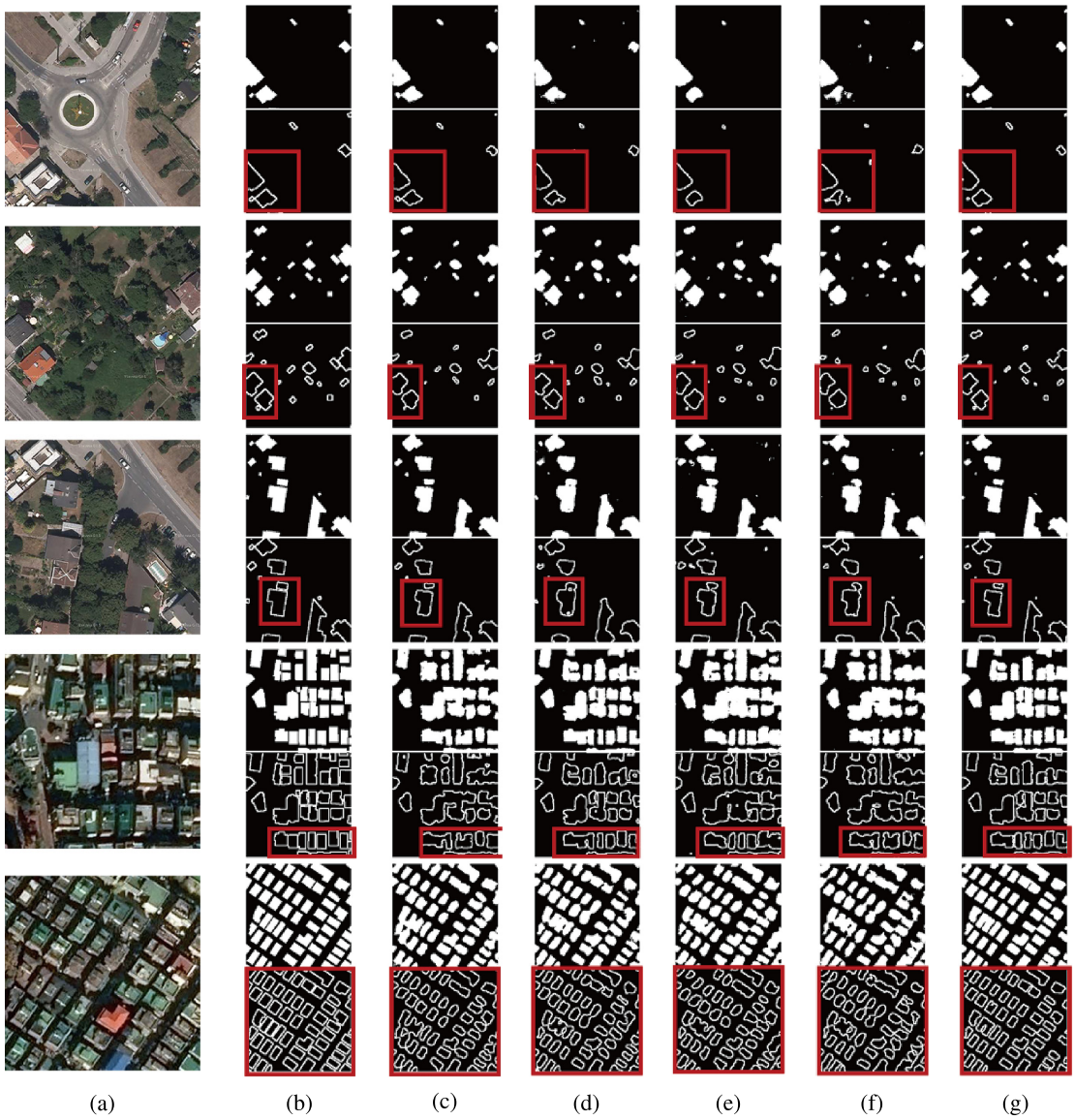


Fig. 14. (Top) Segmentation and (Bottom) boundary detected results by our proposed architecture and other state-of-the-art models. (a) Raw image. (b) Ground truth. (c) Adapted-FCN. (d) U-NetPPL. (e) MC-FCN. (f) MS-CDN. (g) B3SM (ours). Significant changes can be found in areas with red rectangles.

utilized with USIM and B-Loss. USIM outperformed other merge operations with two-scheme learning and offered the higher evaluation scores than other deconvolution operations that are used in one-scheme learning. B-Loss improved the B-IoU in most cases, but the utilization of B-Loss in the second scheme (S_2) reduces an IoU value while increasing an B-IoU value.

Fig. 14 presents segmented results for the B3SM and the other state-of-the-art models discussed above. The B3SM provides enhanced building segmentation with accurate boundaries. In contrast, the prediction results of the other state-of-the-art models contain scattered noise in their backgrounds and unclear building boundaries based on the close proximity of the buildings. In particular, unclear boundaries of buildings are more likely to be observed in the results obtained using the Adapted-FCN and UNetPPL models. These results demonstrate that the B3SM can allow clearly distinguishing the boundaries of segmented buildings by dividing different buildings and can also reconstruct accurate shapes for the segmented boundaries of buildings. These experimental results demonstrate that the B3SM outperforms the other deep learning architectures for building segmentation, resulting in more accurate shapes and boundaries for segmented buildings in aerial images.

VI. Discussion

In this article, we proposed a novel deep learning model with two-scheme learning based on the USIM operator and the B-Loss function. In the first scheme of the proposed architecture, a raw input image is transformed into a prepredicted segmentation map, after which it is refined in the second scheme. Unlike numerous state-of-the-art models, the proposed model is customized to extract precise boundaries and shapes of buildings from aerial images with the high IoU and B-IoU values. In this study, B3SM adopted FusionNet as the baseline architecture since FusionNet has the fundamental

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operations of CNNs, such as concatenation, addition operation, and residual blocks for semantic segmentation [47]. However, the baseline architecture of the B3SM can be any state-of-the-art model. The performance of the B3SM can be further improved when more advanced state-of-the-art architectures are applied to B3SM, which will be examined in future work.

In the building segmentation of aerial images, the use of relatively low-resolution images and consecutive pooling layers in deep learning architecture can result in a loss of object details. To compensate for these missing details, the USIM was proposed. The USIM acts as a merging operation, which is similar to a skip connection in U-Net [53] or FusionNet [47]. In U-Net, skip connections merge two feature maps by concatenating the channels of each feature map. In FusionNet, a deconvolutional block with skip connections merges the channels of two feature maps by using an addition operator for each channel. In contrast, the USIM combines the channels of two input feature maps by performing an upsampling merge operation for each channel in the feature maps. Merged images generated by the USIM are similar to superimposed images generated by blending methods. However, unlike conventional superimposed images, the pixels of the merged images have continuity in their arrangement, meaning that they have different characteristics compared to the pixels in conventional superimposed images [17]. This continuity between pixels can more accurately preserve the detected boundaries of buildings. During the binary classification of buildings, buildings with unclear boundaries degrade segmentation performance, resulting in missing buildings or the merging of separate buildings. However, by using the USIM, the incorrectly predicted pixels among correctly predicted pixels can be effectively corrected. However, since USIM merges the RGB raw input and ground-truth images with different channel information, information confusion might be caused by USIM, while the two images are combined. Nevertheless, as demonstrated in Table VII, USIM showed better performance than the overlay operations, which simply combines two images as one. Note that USIM has distinct advantages, such as preserving the original pixel values of two input images and generating better segmentation maps with clearer boundaries of buildings. The robustness of USIM in terms of information confusion is illustrated in the Supplementary Material.

Furthermore, the B-Loss function provides a clear distinction between the boundaries of buildings during segmentation. Unlike conventional loss functions [26], [27], [54], such as binary cross entropy and L2 loss, B-Loss allows one to obtain accurate shapes of buildings with clear boundaries in final predicted segmentation maps. For the B-Loss, a Canny edge filter was used since a Canny edge detection function is differentiable. Note that every composite function for a loss function needs to be differentiable. Moreover, the nonmaximum-suppression, which is utilized in the Canny edge filter, reduces the amount of computation and increases the mean average precision (mAP) [22], [37]. As a result, unlike other edge detection functions, such as the Sobel edge detection, the Laplacian edge detection, and gradient filters, the Canny edge detection allowed to reduce the training time for B3SM. Therefore, the Canny edge filter was utilized for the B-Loss in B3SM.

The results from the ablation study (see Table VII and Fig. 14) showed that B3SM, with two-scheme learning, USIM, and B-Loss, outperformed other architectures for the segmentation of buildings from aerial images. The combinations of the three proposed new components improved the performance of B3SM, resulting in a better prediction of boundaries and shapes of buildings from aerial images. Hence, B3SM has great potential as novel deep learning architecture to enable improved prediction of the boundaries and shapes of buildings from an aerial image

VII. Conclusion

In this study, we developed a novel deep learning model called the B3SM based on two-scheme learning, USIM, and B-Loss for the enhanced boundary-oriented segmentation of buildings in aerial images. The B3SM allowed us to achieve accurate shapes and boundaries for segmented buildings.

Extensive experiments demonstrated the superior performance of the B3SM compared to state-of-the-art semantic segmentation models. The average IoU and B-IoU values of the B3SM for building segmentation tasks were 76.08% and 78.16%, respectively, which were 8.97% and 8.42% greater than those of the other state-of-the-art deep-learning models, respectively. In the B3SM, two-scheme learning with the USIM and B-Loss significantly enhances the accuracy of building segmentation. Furthermore, it was determined that B-IoU is very helpful for evaluating whether a deep learning model can accurately predict the shapes of buildings, indicating that B-Loss has potential as a novel metric for evaluating the shapes of segmented objects.

Finally, it is important to note that the proposed model can also be applied to the multiclass segmentation of objects, such as roads, trees, and water, with further refinements. Two-scheme learning with the USIM can also enhance the accuracy of the multiclass segmentation of aerial maps, which will be examined in the future.

References

[1] B. C. Reed, J. F. Brown, D. VanderZee, T. R. Loveland, J. W. Merchant, and D. O. Ohlen, "Measuring phenological variability from satellite imagery," *J. Vegetation Sci.*, vol. 5, no. 5, pp. 703–714, Oct. 1994.

[2] G. Wu *et al.*, "Automatic building segmentation of aerial imagery using multi-constraint fully convolutional networks," *Remote Sens.*, vol. 10, no. 3, p. 407, Mar. 2018.

[3] Y. Guo, J. Ji, X. Lu, H. Huo, T. Fang, and D. Li, "Global-local attention network for aerial scene classification," *IEEE Access*, vol. 7, pp. 67200–67212, 2019.

[4] J. Blundell and D. Opitz, "Object recognition and feature extraction from imagery: The feature analyst approach," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 36, no. 4, p. C42, 2006.

[5] X. Huang and L. Zhang, "Road centreline extraction from high-resolution imagery based on multiscale structural features and support vector machines," *Int. J. Remote Sens.*, vol. 30, no. 8, pp. 1977–1987, Apr. 2009.

[6] M. Mokhtarzade and M. J. V. Zoej, "Road detection from high-resolution satellite images using artificial neural networks," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 9, no. 1, pp. 32–40, Feb. 2007.

[7] A. P. D. Poz, R. A. B. Gallis, J. F. C. da Silva, and É. F. O. Martins, "Object-space road extraction in rural areas using stereoscopic aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 4, pp. 654–658, Jul. 2012.

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[8] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.

[9] O. Oktay *et al.*, "Attention U-Net: Learning where to look for the pancreas," 2018, *arXiv:1804.03999*. [Online]. Available: http://arxiv.org/abs/1804.03999

[10] X. Wang, A. Shrivastava, and A. Gupta, "A-fast-RCNN: Hard positive generation via adversary for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2606–2615.

[11] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[12] G. Cheng, Z. Li, J. Han, X. Yao, and L. Guo, "Exploring hierarchical convolutional features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6712–6722, Nov. 2018.

[13] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Springer, 2015, pp. 234–241.

[14] J. H. Kim *et al.*, "Objects segmentation from high-resolution aerial images using U-Net with pyramid pooling layers," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 115–119, Jan. 2019.

[15] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[16] F. Alidoost, H. Arefi, and F. Tombari, "Building outline extraction from aerial images using convolutional neural networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vols. XLII–4/W18, pp. 57–61, Oct. 2019.

[17] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The Inria aerial image labeling benchmark," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3226–3229.

[18] E. Maggiori, G. Charpiat, Y. Tarabalka, and P. Alliez, "Recurrent neural networks to correct satellite image classification maps," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 4962–4971, Sep. 2017.

[19] L. Mou and X. X. Zhu, "RiFCN: Recurrent network in fully convolutional network for semantic segmentation of high resolution remote sensing images," 2018, *arXiv:1805.02091*. [Online]. Available: http://arxiv.org/abs/1805.02091

[20] X. Li, X. Yao, and Y. Fang, "Building-A-Nets: Robust building extraction from high-resolution remote sensing images with adversarial networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3680–3687, Oct. 2018.

[21] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.

[22] T. Hu, Y. Wang, Y. Chen, P. Lu, H. Wang, and G. Wang, "Sobel heuristic kernel for aerial semantic segmentation," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3074–3078.

[23] P. Kaiser, J. D. Wegner, A. Lucchi, M. Jaggi, T. Hofmann, and K. Schindler, "Learning aerial image segmentation from online maps," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6054–6068, Nov. 2017.

[24] M. Volpi and D. Tuia, "Dense semantic labeling of subdecimeter resolution images with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 881–893, Feb. 2017.

[25] K. Yue, L. Yang, R. Li, W. Hu, F. Zhang, and W. Li, "TreeUNet: Adaptive tree convolutional neural networks for subdecimeter aerial image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 156, pp. 1–13, Oct. 2019.

[26] K. Doi and A. Iwasaki, "The effect of focal loss in semantic segmentation of high resolution aerial image," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 6919–6922.

[27] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[28] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "High-resolution aerial image labeling with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7092–7103, Dec. 2017.

[29] M. Volpi and D. Tuia, "Deep multi-task learning for a geographically-regularized semantic segmentation of aerial images," *ISPRS J. Photogramm. Remote Sens.*, vol. 144, pp. 48–60, Oct. 2018.

[30] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.

[31] S. Saha, L. Mou, C. Qiu, X. X. Zhu, F. Bovolo, and L. Bruzzone, "Unsupervised deep joint segmentation of multitemporal high-resolution images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8780–8792, Dec. 2020.

[32] D. Marmanis, K. Schindler, J. D. Wegner, S. Galliani, M. Datcu, and U. Stilla, "Classification with an edge: Improving semantic image segmentation with boundary detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 158–172, Jan. 2018.

[33] E. Fernandez-Moral, R. Martins, D. Wolf, and P. Rives, "A new metric for evaluating semantic segmentation: Leveraging global and contour accuracy," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2018, pp. 1051–1056.

[34] F. Liu, G. Lin, and C. Shen, "CRF learning with CNN features for image segmentation," *Pattern Recognit.*, vol. 48, no. 10, pp. 2983–2992, Oct. 2015.

[35] S. Wu, J. Yang, X. Wang, and X. Li, "IoU-balanced loss functions for single-stage object detection," 2019, *arXiv:1908.05641*. [Online]. Available: http://arxiv.org/abs/1908.05641

[36] F. van Beers, A. Lindström, E. Okafor, and M. Wiering, "Deep neural networks with intersection over union loss for binary image segmentation," in *Proc. 8th Int. Conf. Pattern Recognit. Appl. Methods*, 2019, pp. 438–445.

[37] X. Hu, Y. Liu, K. Wang, and B. Ren, "Learning hybrid convolutional features for edge detection," *Neurocomputing*, vol. 313, pp. 377–385, Nov. 2018.

[38] K. Kim and S. Young Chun, "SREdgeNet: Edge enhanced single image super resolution using dense edge detection network and feature merge network," 2018, *arXiv:1812.07174*. [Online]. Available: http://arxiv.org/abs/1812.07174

[39] R. Kohavi *et al.*, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in *Proc. Int. Joint Conf. AI*, vol. 14. no. 2, Aug. 1995, pp. 1137–1145.

[40] G. A. Baxes, *Digital Image Processing: Principles and Applications*. Hoboken, NJ, USA: Wiley, 1994.

[41] M. Abadi *et al.*, "Tensorflow: A system for large-scale machine learning," in *Proc. 12th USENIX Symp. Operating Syst. Design Implement. (OSDI)*, 2016, pp. 265–283.

[42] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*. [Online]. Available: http://arxiv.org/abs/1412.6980

[43] L. Bottou, "Large-scale machine learning with stochastic gradient descent," in *Proc. COMPSTAT*. Springer, 2010, pp. 177–186.

[44] Y. Wu and K. He, "Group normalization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.

[45] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*. [Online]. Available: http://arxiv.org/abs/1502.03167

[46] G. Csurka, D. Larlus, and F. Perronnin, "What is a good evaluation measure for semantic segmentation?" in *Proc. Brit. Mach. Vis. Conf.*, 2013, p. 2013.

[47] T. Minh Quan, D. G. C. Hildebrand, and W.-K. Jeong, "FusionNet: A deep fully residual convolutional neural network for image segmentation in connectomics," 2016, *arXiv:1612.05360*. [Online]. Available: http://arxiv.org/abs/1612.05360

[48] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[49] Z. Wang, E. P. Simoncelli, and A. C. Bovik, "Multiscale structural similarity for image quality assessment," in *Proc. 37th Asilomar Conf. Signals, Syst. Comput.*, Nov. 2003, pp. 1398–1402.

[50] A. Rehman, Y. Gao, J. Wang, and Z. Wang, "Image classification based on complex wavelet structural similarity," *Signal Process., Image Commun.*, vol. 28, no. 8, pp. 984–992, Sep. 2013.

[51] K. I. Gandhi, S. Janarthanan, R. Sathish, and A. Surendar, "Dominant feature prediction by improved structural similarity computation," in *Proc. Int. Conf. Innov. Trends Inf. Technol. (ICITIIT)*, Feb. 2020, pp. 1–5.

[52] T. Kobayashi, "Structured feature similarity with explicit feature map," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1211–1219.

[53] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 3431–3440.

[54] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, Apr. 1960.

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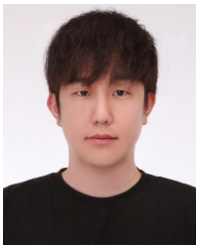
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